Solution to Assignment 2:-

[1] ① $\rho(\phi) = a\mu(\phi) + b\nu(\phi) = 0$

② For $A_i \in \mathcal{A}$, $i = 1, 2, \cdots$, are pairwise disjoint, we have

$$\rho\left(\bigcup_{i=1}^{\infty} A_i\right) = a\mu\left(\bigcup_{i=1}^{\infty} A_i\right) + b\nu\left(\bigcup_{i=1}^{\infty} A_i\right)$$
$$= \sum_{i=1}^{\infty} a\mu(A_i) + \sum_{i=1}^{\infty} b\nu(A_i)$$
$$= \sum_{i=1}^{\infty} \rho(A_i).$$

Hence ρ is a measure on $(X, \mathcal{A})$.

[2] Since $\rho(A_j) = 1$, we obtain $\rho(A_j^c) = 0$.

By proposition 1.1.17,

$$\rho\left(\bigcup_{j=1}^{\infty} A_j^c\right) \le \sum_{j=1}^{\infty} \rho(A_j^c) = 0$$

Hence, by De Morgan Laws,

$$\rho\left(\bigcap_{j=1}^{\infty} A_j\right) = \rho\left(\left(\bigcup_{j=1}^{\infty} A_j^c\right)^c\right) = 1$$

[3] Since μ is a finite measure, by Prop. 1.1.17(3), we get

$$\mu\left(\bigcup_{j=1}^{\infty} A_j\right) - \mu\left(\bigcup_{j=1}^{\infty} B_j\right) = \mu\left(\left(\bigcup_{j=1}^{\infty} A_j\right) \setminus \left(\bigcup_{j=1}^{\infty} B_j\right)\right)$$
$$= \mu\left(\left(\bigcup_{j=1}^{\infty} A_j\right) \cap \left(\bigcup_{j=1}^{\infty} B_j\right)^c\right)$$
$$= \mu\left(\bigcup_{j=1}^{\infty} \left(A_j \cap \left(\bigcup_{j=1}^{\infty} B_j\right)^c\right)\right)$$
$$\le \sum_{j=1}^{\infty} \mu\left(A_j \cap \left(\bigcup_{j=1}^{\infty} B_j\right)^c\right)$$

But $A_j \cap \left(\bigcup_{j=1}^{\infty} B_j\right)^c = A_j \cap \left(\bigcap_{j=1}^{\infty} B_j^c\right) \subset A_j \cap B_j^c = A_j \setminus B_j$.

So $\mu\left(\bigcup A_i\right) - \mu\left(\bigcup B_i\right) \le \sum \mu(A_i \setminus B_i) = \sum \mu(A_i) - \mu(B_i)$

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[4] We only need to prove countably additive property for μ.

First we show that $\mu(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i)$, for pairwise disjoint sets $A_1, A_2, \cdots A_n$.

$$\mu(\bigcup_{i=1}^{n} A_i) = \mu(A_1 \cup (\bigcup_{i=2}^{n} A_i)) = \mu(A_1) + \mu(\bigcup_{i=2}^{n} A_i)$$

$$= \sum_{i=1}^{n} \mu(A_i).$$

Notice that $\bigcup_{i=1}^{n} A_i \nearrow \bigcup_{i=1}^{\infty} A_i$ as $n \to +\infty$, we have

$$\mu(\bigcup_{i=1}^{\infty} A_i) = \lim_{n\to\infty} \mu(\bigcup_{i=1}^{n} A_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mu(A_i) = \sum_{i=1}^{\infty} \mu(A_i)$$

Hence, μ is a measure on $(\Omega, \mathcal{A})$.

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